Skills Summary

Matching Division Stories, Drawings, and Equations

Use this reference while completing your worksheet or when studying.

Every division has the same three parts.

- the **total** — how many there are in all
- the **number of groups**
- the **size of each group** — how many are in one group

A story, a drawing, and an equation all tell those same three parts. Two of them are always given; the third is what you find.

total $\div$ number of groups $=$ size of each group   total $\div$ size of each group $=$ number of groups

1. From a story to an equation

Find the **total** first — it is the number you divide. Then ask what the story gives you next: the number of groups, or how many go in each group. The part it never tells you is the answer.

Example: 15 grapes are shared equally among 3 children. How many does each child get?
Step 1. The total is 15 and the story names 3 groups, so the missing part is the size of one group — that is the sharing shape of division.
Step 2. Write total $\div$ groups and share them out: $15 \div 3$ puts 5 grapes in each pile.

$$15 \div 3 = \mathbf{5}$$

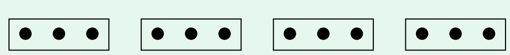
Try it: 21 marbles are shared equally among 3 jars. How many marbles are in each jar?

check: 7

2. From a drawing to an equation

A drawing hands you two parts for free: **count the dots** for the total, and **count the groups** (boxes, circles, or rows). Then write total $\div$ groups. An array works the same way — rows are the groups.

Example: Write the division equation this picture shows.



Step 1. The picture is already grouped, so read the two given parts off it rather than counting one dot at a time.
Step 2. There are 12 dots in all and 4 boxes, so the equation is $12 \div 4$, and one box holds 3.

$$12 \div 4 = \mathbf{3}$$

Try it: A picture shows 16 dots sorted into 2 equal boxes. Write the division equation and answer it.

check: **8**

3. From an equation back to a story

Say the equation out loud: "*this many* in all, split into *this many* equal parts." The number in front of the $\div$ sign must be the **total** in the story. If a story uses that number as "how many in each," it does not match.

Example: Which story matches $20 \div 5$? **A.** 20 books put on 5 equal shelves. **B.** 20 books on each of 5 shelves.
Step 1. The number written first must be the total, so check which story uses 20 as the whole amount.
Step 2. Story A uses 20 as the total and 5 as the number of groups, so A matches; in B the total would be much larger than 20. Answering A gives 4 books on a shelf.

$$20 \div 5 = \mathbf{4}$$

Try it: Write a sharing story for $30 \div 5$, then answer it.

check: **6**

4. Fact families: multiply to check

Three numbers make four sentences. From $4 \times 6 = 24$ you also get $6 \times 4 = 24$, $24 \div 4 = 6$ and $24 \div 6 = 4$.

So a missing number in a division sentence can always be found by turning the sentence into multiplication and skip-counting.

Example: Find the missing number: $4 \times \square = 24$.
Step 1. A missing factor is what division finds, so turn the sentence around instead of guessing numbers.
Step 2. The partner sentence is $24 \div 4$; skip-count by fours — 4, 8, 12, 16, 20, 24 — which takes 6 steps.

$$\square = \mathbf{6}$$

Try it: Find the missing number: $8 \times \square = 56$.

check: **7**

(!) Watch out: the total is the number you divide, and it is never the answer to a sharing question — if your answer is bigger than the total, you multiplied. And read what the question asks for: in an array, the number of rows and the number in each row are different answers.